***EE/CprE/SE 492 WEEKLY REPORT 4***

***October 2, 2025 - October 16, 2025***

***Group number:*** *16*

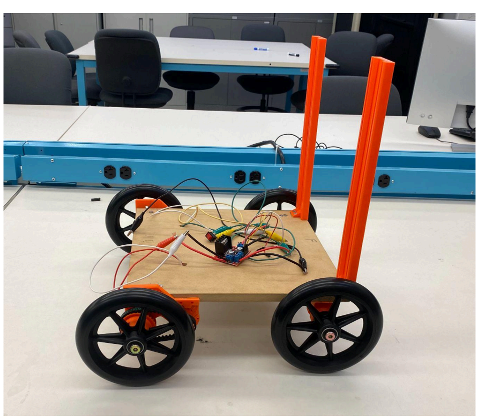
***Project title:*** *Outdoor Roomba*

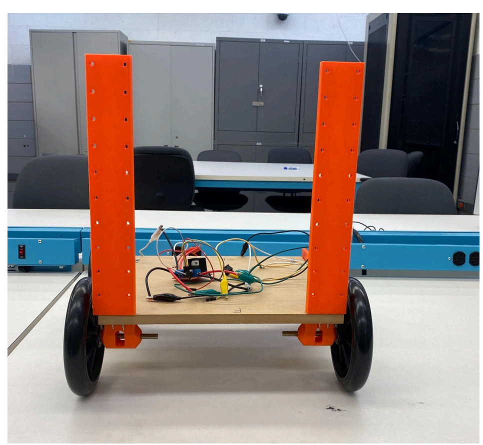
***Client &/Advisor:*** *Prof. Diane Rover*

***Team Members/Role:***
*Brodie M Bates -* ***Embedded system***
*Daniel Vergara Pinilla -* ***Machine Learning***
*Mitchell D Owen -* ***Project Tester***
*Om Shukla -* ***Frontend Developer***
*Sudipta Halder -* ***Simulation developer***

- **Weekly Summary**

The first week on the hardware side we had the front tires installed, solved our battery problems, tested the robot wirelessly with a raspberry pi via wifi on ssh. This week we worked on the LiDAR sensor, cut some boards for the walls of the robot and installed the brackets for the arm.





On the software side: We successfully deployed the website for the User UI/UX, made some changes to the software to fix the issue that the robot constantly takes pictures, but instead it will take a proctor every 2s and then analyzes the picture to detect trash, power saving and more efficient for the AI model

- **<u>Past weeks accomplishments</u>**

  - Sudipta Halder: Worked on redesigning the shovel to account for the scuttle robot's wheels and motos. Made the arms thicker to remove any bending.

  - Brodie Bates: Got LiDAR sensor working and reporting data, tested the battery pack, wrote the code to run the motor controllers.

  - Daniel Pinilla: Deployed the website on the Server, and worked on changes to the main code so the robot will not constantly take pictures constantly but every 2s to reduce repetition and improve battery and time performance

  - Om Shukla: Started writing and testing endpoints and server connections from frontend to backend since the website was uploaded to the server.

  - Mitchell Owen: Tested battery pack and driving mechanisms. Making a list of parts to upgrade the chassis of the robot.

- **Pending issues**

Sudipta Halder: Classes and other work have been interfering with work on the robot for the past week. The plan is to finish the other work outside of the senior design project and complete the hardware portion of this project so we can get started on implementation and user testing soon.

- **Individual contributions**

| **NAME** | **Individual Contributions** | **Hours this week** | **HOURS cumulative** |
| --- | --- | --- | --- |
| Brodie Bates | LiDAR sensor code, battery troubleshooting and testing. | 16 | 39 |
| Om Shukla | Creating and testing endpoints for server connection. | 4 | 21 |
| Sudipta Halder | Redesigning the parts in Fusion 360, programmed a raspberry pi to make the robot be controlled wirelessly. | 12 | 43 |
| Daniel Mauricio Vergara | Deploying the website and working on changes to the main Pi code to change the main workflow of the robot for more efficient management | 14 | 30 |
| Mitchell Owen | continue to assemble and test the chassis of the robot and provide support to Brodie and Sudipta on the battery and motor application. | 10 | 23 |

- **Plans for the upcoming week**

  - Sudipta Halder: Finish work on arm and the power shovel so we can start adding the sensors and microcontroller and move on to implementation.
  - Brodie Bates: This week I'll be working with the ping sensors for the lower portion of the robot as well as getting the data out of the LiDAR formatted better and faster.
  - Om Shukla: Finish login endpoints and start working on endpoints to retrieve roomba information from the server.
  - Mitchell Owen: Finish the design on the robot and assemble side walls and "power shovel". Also make compartments for the batteries and other electrical components for the motors.
  - Daniel Pinilla: This week I will strt testing the connection and the runtime of all code together, including the orientation and camera capture as well as mocking the robot movement commands

- **Summary of weekly advisor meeting**

In the advisor meeting we demoed the LiDAR code and talked about our sensor placement. We also talked more about our design and potential flaws that could come from having a ramp on the inside of the robot. This could lead to trash being jammed in places we do not want. We were advised to brainstorm some ideas for trash dumping after the roomba has picked up all objects.

Next, we showed a diagram of how our robot will work and Dr. Rover had some questions for us about our interfaces and would be more interested to see how those are working, the ones mentioned were between the pi and motor controllers. Another problem with our diagram is it should show physical connections and include wireless connections in our system. Lastly, we should show the frontend and backend better as well as what kind of communication interface is used on the message queue.